

Zirui Zhao

Curriculum Vitae

28 Xianning West Road
Xi'an, Shaanxi, China, 710049
☎ +86 159 2995 3178
✉ ryan_zzr@outlook.com
🌐 www.ryanzhao.org
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Education

- 2020 - 2024 **Ph.D.**, *Computer Science*, School of Computing, National University of Singapore, Singapore.
2016 - 2020 **B.E.**, *Automation*, Faculty of Elec. & Info. Engineering, Xi'an Jiaotong University, Xi'an, China.

Visiting

- 2019 Summer **Summer intern**, *Safe AI Lab*, Carnegie Mellon University, Pittsburgh, USA.
2018 Summer **Visiting student**, *School of Computing*, National University of Singapore, Singapore.

Publications & Manuscripts

- [2] R. Chen, W. Wang, **Z. Zhao**, D. Zhao, "Active Learning for Risk-Sensitive Inverse Reinforcement Learning," submitted to ICRA 2020, preprinted in arXiv.
[1] **Z. Zhao**, Y. Mao, Y. Ding, P. Ren and N. Zheng, "Visual-Based Semantic SLAM with Landmarks for Large-Scale Outdoor Environment," 2019 2nd China Symposium on Cognitive Computing and Hybrid Intelligence (CCHI), Xi'an, China, 2019, pp. 149-154.

Selected Researches & Projects

- 2019 Summer **Active Risk-sensitive Inverse Reinforcement Learning**, *Safe AI Lab*, CMU.
Risk-sensitive inverse reinforcement learning provides an general model to capture how human assess the distribution of a stochastic outcome when the true distribution is unknown (ambiguous). This work enables an RS-IRL learner to actively query expert demonstrations for faster risk envelope approximation.
Nov. 2018 - **Visual Semantic SLAM**, *Cognitive Architecture Lab*, XJTU.
Jun. 2019 This work combines the semantic labels with pointclouds generated from ORB-SLAM by using bayesian update rules. Besides, the landmarks in the real world is attached in the map, associated with the pointclouds by using fuzzy mathematics.
Apr. 2018 - **Multi-robot Cooperative navigation**, *Cognitive Architecture Lab*, XJTU.
Mar. 2019 This project established a multi-robot navigation and exploration system, which consist of UAVs and UGVs.
Demo video link: <https://www.youtube.com/watch?v=wSB4hyW9rWc&t=5s>

Selected Scholarships & Honors

- 2020 - 2024 NUS Research Scholarship
2017 Siyuan Merit Scholarship
2018 2nd Prize of 1989 Mechanical Engineering Alumni Scholarship for Qian Xue Sen Program
2019|2018|2017 Excellent Student title
2020 Excellent Graduate title

Interests & Skills

- Interests Cognitive Robotics, Robot Planning under Uncertainty, Probabilistic Robotics, Human-Robot Interaction.
Language C++, Python, MATLAB, $\LaTeX$
Tools TensorFlow, PyTorch, Keras, Scikit-Learn, Numpy, Jupyter, OpenCV, ROS.
Software Carla, GitLab, GitHub.